Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2024

Bachelor of Computer Applications SEMESTER - III

Course Code: BCA202-3 **Max Marks: 100**

Course Name: OPERATING SYSTEM **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Demonstrate an understanding of the basic working process of an operating system.

CO2: Explain the importance of process and scheduling.

CO3: Analyse the issues in synchronization and memory management.

Section A **2*10=20**

1)	What is a Kernel?	**2**	(CO1, L2)
2)	Illustrate a system call with an example.	**2**	(CO1, L3)
3)	What is the turnaround time of a process? How is it calculated?	**2**	(CO1, L2)
4)	Explain the difference between arrival time and burst time of a process.	**2**	(CO2, L4)
5)	Relate the difference between deadlock prevention and deadlock avoidance.	**2**	(CO2, L3)
6)	What is a deadlock in the context of operating systems?	**2**	(CO2, L4)
7)	What are the hardware functions required to support demand paging.	**2**	(CO2, L2)
8)	What is a file allocation table?	**2**	(CO3, L5)
9)	What is meant by free-space management in file systems?	**2**	(CO3, L5)
10)	Define rotational latency.	**2**	(CO3, L4)

Section B **5*6=30**

11)	Explain Process Control Block and explain its parameters.	**5**	(CO1, L1)
12) a)	Race conditions are possible in many computer systems. Consider a banking system that maintains an account balance with two functions: deposit(amount) and withdraw(amount). These two functions are passed the amount that is to be deposited or withdrawn from the bank account balance. Assume that a husband and wife share a bank account. Concurrently, the husband calls the withdraw() function, and the wife calls deposit(). Describe how a race condition is possible and what might be done to prevent the race condition from occurring.	**5**	(CO2, L3)
	OR		
b)	Explain wait() and signal() operations in semaphore.	**5**	(CO2, L3)
13)	Compare and contrast how deadlock prevention differs from deadlock avoidance.	**5**	(CO1, L2)
14) a)	What is thrashing? How is it controlled by OS?	**5**	(CO3, L5)

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OR

b)	What is the purpose of a directory in a file system?Explain.	**5**	(CO3, L5)
15)	Illustrate the states of a process. Explain with a neat diagram.	**5**	(CO3, L3)
16)	Explain how the LOOK and C-LOOK disk scheduling algorithms work.	**5**	(CO1, L2)

Section C **10*5=50**

17) a)	Illustrate the working principle of an Interrupt with a neat diagram.	**10**	(CO2, L3)

OR

b)	Consider the processes P1, P2, P3 given in the below table, arrives for execution in the same order, with Arrival Time 0, and given Burst Time,	**10**	(CO2, L3)

Process	Arrival Time	Burst Time
P1	0	24
P2	0	3
P3	0	3

a) Draw Gantt charts illustrating the execution of these processes using the following scheduling algorithms:

i) Shortest Job First (SJF)

ii) Round Robin (Quantum = 3 ms)

b) Calculate the average waiting time and average turnaround time for each algorithm.

c) Which algorithm gives the best average waiting time for this set of processes?

18) a)	Examine the following questions using the banker's algorithm: Given resources A has 7 instances,B has 2 instances and C has 6 instances	**10**	(CO3, L4)

	Allocation			Request			Available		
P0	0	1	0	0	0	0	0	0	0
P1	2	0	0	2	0	2			
P2	3	0	3	0	0	0			
P3	2	1	1	1	0	0			
P4	0	0	2	0	0	2			

a. Is the system in a safe state?

b. If a request from process P2 arrives for (0,0,1), can the request be granted immediately?

OR

b)	For the Page Reference String:7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0. 1. 7, 0, 1Calculate the Page Faults applying (i)Optimal(ii)LRU and(iii)FIFO PageReplacement Algorithms for a Memory with three frames.	**10**	(CO3, L4)
19) a)	Explain the various types of disk scheduling algorithms (FCFS, SSTF, C-SCAN, LOOK) with a suitable example.	**10**	(CO2, L3)

OR

b)	Describe in detail the process of file system implementation, including the file control block (FCB) and directory structure.	**10**	(CO2, L3)
20) a)	Categorize the impact of different memory allocation strategies on system performance.	**5**	(CO2, L2)
b)	Compare and contrast First-Come First-Served (FCFS), Shortest Job First (SJF) scheduling algorithms.	**5**	(CO2, L2)
21)	Discuss the concepts of file fragmentation and defragmentation in file systems.	**10**	(CO3, L3)

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Revised Bloom's Taxonomy (RBT) Levels		
L1 - Remember	L2 - Understand	L3 - Apply
L4 - Analyse	L5 - Evaluate	L6 - Create

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